

Computer Vision

CSE5CV

LMS Subject link

<https://lms.latrobe.edu.au/course/view.php?id=137262>

SUBJECT LEARNING GUIDE

Teaching Period Semester 2, 2023

Bundoora

Subject Coordinator

Dr Aiden Nibali

SUBJECT DETAILS

Subject Code:	CSE5CV
Subject Title:	Computer Vision
Teaching Period:	Semester 2
Location(s):	Bundoora
Credit Points:	15
Delivery Mode:	Daytime
AQF Level:	Level 9 – Masters Degree

ENROLMENT REQUIREMENTS

Prerequisites:	CSE4IP or CSE4002
Co-requisites:	None
Assumed Skills & Knowledge:	Basic programming skills
Special Study Requirements:	None

Additional subject requirements:

None.

Class materials:

Students are expected to have access to a desktop or laptop computer with Internet access.

STAFF CONTACTS

Subject Coordinator:	Dr Aiden Nibali
Email:	a.nibali@latrobe.edu.au
Tel:	-
Location:	Room 220, Physical Sciences 1, Bundoora Campus
Consultation time/s:	By appointment

ABOUT THIS SUBJECT

Computer vision is an interdisciplinary scientific field that deals with how computers can be made to gain high-level understanding from digital images or videos. From the perspective of engineering, it seeks to automate tasks that the human visual system can do. In this subject, you will be introduced to topics in computer vision, covering from early vision to mid and high-level vision such as camera imaging geometry, feature detection and matching, stereo, motion estimation and tracking, scene understanding and image captioning. You will practice statistical models and machine learning models for various computer vision tasks. You will have the opportunity to implement algorithms for real-world computer vision applications in labs.

SUBJECT INTENDED LEARNING OUTCOMES (SILOs)

Upon successful completion of this subject, you should be able to:

- 1 Critique image processing techniques for basic computer vision tasks and address design issues.
- 2 Implement and analyse statistical and machine learning models for advanced computer vision tasks and address design issues.
- 3 Design, implement and evaluate a machine vision model to recognise visual concepts from images.
- 4 Analyze a given real-world problem and investigate different computer vision approaches to provide an optimal solution.

GRADUATE CAPABILITIES (GC)

GC	GC Point
Communication	N/A
Inquiry and Analysis	N/A
Personal and Professional	N/A
Discipline Knowledge and Skills	N/A

LEARNING ACTIVITIES

Timetable details are available at <https://www.latrobe.edu.au/students/your-course/timetables>. Students should also regularly check this subject's LMS site for updated information.

Week	Learning Activities	Lecture topics	Contact hours
1	Lecture	Introduction to computer vision, digital image fundamentals	2
2	Lecture, lab	Image features and histograms, contrast enhancement, denoising and filtering operations	4
3	Lecture, lab	Image feature extraction, feature-based classification models, classification performance evaluation	4
4	Lecture, lab	Convolutional neural network classification models	4
5	Lecture, lab	Object detection and recognition	4
6	Lecture, lab	Face detection and recognition	4
7	Lecture, lab	Video classification and human pose estimation	4
8	Lecture, lab	Synthetic image generation	4
9	Lecture, lab	Microsoft Azure for computer vision	4
10	Lecture, lab	Computer vision applications: virtual and augmented reality	4
11	Lecture, lab	Computer vision applications: multimedia and computer graphics	4
12	Lecture	Revision	2
		Total	44

Details of any self-allocation activities

The intro lab (Lab 00) is to be completed in advance of the first scheduled lab class.

Students are to complete unfinished labs in their own time. Students are to complete assessment tasks in their own time (observing the due dates).

SELF-DIRECTED STUDY

In addition to the learning activities specified, you are expected to allocate approximately **6 hours per week of self-directed study in this subject**. This time includes subject preparation, completing learning activities, readings, assessment preparation and other related work to enhance your participation in this subject and your learning at La Trobe.

LEARNING RESOURCES & TECHNOLOGIES

Resource Type	Title	Author and Year	Publisher
Recommended (not required)			
Book	Computer Vision Algorithms and Applications	Richard Szeliski, 2021	Springer

ASSESSMENT AND FEEDBACK SUMMARY

Note: all assessment instructions and rubrics with detailed criteria and level of performance are found on the Learning Management System two weeks prior to classes commencing.

Week	Date due	%	Assessment type	Key Criteria	Feedback method	SILOs assessed
2	25 August	10	Individual assignment	Length equivalent to 500 words	Comprehensive mark breakdown within three weeks	1, 2
8	20 October	40	Individual assignment	- Length equivalent to 2000 words - One-on-one demonstration of work	Comprehensive mark breakdown within three weeks, feedback during demo	3, 4
Final assessment period	TBA	50	2-hour final examination	Length equivalent to 2000 words	Final mark as reported by the university	1, 2, 3, 4

Additional information about assessment & feedback

Students should keep a copy of all assignments submitted for assessment.

MANDATORY REQUIREMENTS

There are a variety of assessment hurdles or conditions for passing a subject other than the overall mark. **Mandatory Requirements** may be built into the assessment of the learning outcomes of your subject.

There are no Hurdle Requirements for this subject

GENERAL ASSESSMENT REQUIREMENTS

SUBMITTING WRITTEN WORK WITH TURNITIN

Turnitin is a web-based text-matching software system used at La Trobe University to assist you in writing your assignments and checking for similarity with existing published work. Please remember to allow adequate time to submit your assignment to Turnitin. **It is your responsibility to have your assignment submitted by the assessment due date.** Not receiving an originality report is not an acceptable reason for requesting an assignment extension unless it is a required part of the assessment.

POLICIES, PROCEDURES AND GUIDELINES

The University has a comprehensive policy framework to which both staff and students must adhere. You should familiarise yourself with those policies, procedures and guidelines likely to affect you, especially the following:

- [Academic Progress Review Policy](#)
- [Assessment Policy](#)
- [Health and Safety](#) (OHS)
- [Privacy](#)
- [Student Academic Misconduct Policy](#)

ADJUSTMENTS TO ASSESSMENT

You may be affected by a range of adverse circumstances while you are preparing for or undertaking an assessment task. There are avenues for adjustments to your assessment depending on the circumstances you face including opportunities to apply for extensions to submission dates, Special Consideration and Alternative Assessment Schedules. For more detailed information refer to the [Assessment Procedure – Adjustments](#).

REQUEST AN EXTENSION OF TIME TO SUBMIT AN ASSIGNMENT TASK

Where you know in advance that you will not be able to submit an assessment task by the due date due to adverse circumstances that have affected you during the preparation of the task, you will need to request an extension of time to submit. **This must be done at least three days prior to the due date.** Penalties normally apply if you submit an assessment after the due date. **Poor time management is not an acceptable reason for an extension.**

To apply, go to <https://www.latrobe.edu.au/students/admin/forms/request-an-extension/request>

SPECIAL CONSIDERATION

If you have experienced serious short term, adverse and unforeseen circumstances that substantially affect your ability to complete an assessment task to the best of your potential, you may be eligible to apply for **Special Consideration**. To do this or find more information, go to <http://www.latrobe.edu.au/special-consideration>

STUDENTS WITH A LEARNING ACCESS PLAN (LAP)

Students who have a Learning Access Plan must apply for any adjustments they may need via the same channels but may have access to more flexible arrangements depending on their circumstances. Please refer to the [Assessment Procedure – Adjustments](#) and your LAP.

REVIEW AND RE-MARK

For information on what to do if you wish to seek a review of an assessment outcome see Part B of the Validation and Moderation Procedure.

ACADEMIC INTEGRITY

Academic integrity means being honest in academic work and taking responsibility for learning the conventions of scholarship. Academic integrity education is integral to the learning experience at La Trobe University. The University requires its academic staff and students to observe the highest ethical standards in all aspects of academic work, and it demonstrates its commitment to these values by awarding due credit for honestly conducted scholarly work, and by penalising academic misconduct and all forms of cheating.

The penalty for submitting an assignment under your name that is the work of a third party may be severe, even leading to exclusion from the University without readmission. Refer to the **Academic Integrity - Schedule of Penalties and Actions** within the [Student Academic Misconduct Policy](#).

You should understand and be able to practice integrity in all your academic work, be familiar with the [Academic Integrity website](#) and complete the **Academic Integrity Module (AIM)** found in your LMS.

If you have any specific questions about **academic integrity** and your assessment, your Subject or Course Coordinator will be able to assist.

STUDENT FEEDBACK ON SUBJECT SURVEYS

The **Student Feedback on Subjects (SFS) Survey** is one of the quality assurance processes that occur across the University. In this survey, you can tell us about your learning experiences in this subject. We value your feedback. Your views are taken seriously and assist us to improve the learning experience for future students through the LTU Subject Monitoring process.

The surveys will be available for you to complete towards the end of the subject teaching period. Feedback from previous students who completed this subject is available in the LMS site for this subject.

For more information you may contact surveys@latrobe.edu.au.

CHARTER OF STUDENT RIGHTS AND RESPONSIBILITIES

As a La Trobe coursework student, you have rights and responsibilities which are governed by La Trobe statutes and regulations, as well as policies and procedures. La Trobe University aspires to provide an outstanding student experience which is conducive to scholarly activity and supportive of your learning goals.

The **Charter of Student Rights and Responsibilities** is a summary of your rights and responsibilities as a student at La Trobe. The Charter was developed in consultation with La Trobe students.

<https://policies.latrobe.edu.au/document/view.php?id=225>.

STUDENT COMPLAINTS

The **Student Complaints Management System (SCM)** is a formal mechanism for students to lodge concerns about any aspect of their university experience. A student complaint may be about an administrative matter, a problem with a subject or course, or the conduct of a student or staff member.

For more information about the **Student Complaint process** and how to lodge a complaint, visit the **Complaints and Feedback** page at: <https://www.latrobe.edu.au/students/support/complaints>.

GETTING HELP AND SUPPORT AT UNI

A range of services at La Trobe can assist you with your studies, health and wellbeing.

For the full range of services and facilities available for you, go to **Student Help and Support** at <http://www.latrobe.edu.au/students/support>.

STUDYING ONLINE

When you are in studying an online subject, please be sure that you can continue to access our learning support services through the Learning Hub (academic skills, English language, Maths and Science) via Zoom appointments that you can book by accessing the [Learning Hub LMS page](#) or by emailing the LearningHub@latrobe.edu.au.

You also have access to assignment feedback and online study support chat through Studiosity that can be accessed from any subject [LMS](#) page. You can also access the Library services remotely through [the library website](#).

LIBRARY

Your **Library** offers many ways to support your study. To discover more about how the Library can help, visit your **Library website** and explore the following essential resources:

- **Achieve@Uni**
<https://www.latrobe.edu.au/students/study-resources/learning/achieve> - Learn skills to help complete your assessments
- **Assessment Planner**
<http://www.latrobe.edu.au/library/assessment-thesis-support/plan-my-assessment> - Create a personal assessment schedule
- **Expert Help Guides**
<https://latrobe.libguides.com/> - Find expert discipline and specialist help for your research
- **Referencing at La Trobe University**
<http://latrobe.libguides.com/referencing/> - Understand how and why to reference
- **Academic Referencing Tool**
www.lib.latrobe.edu.au/referencing-tool - Get detailed examples of referencing styles
- **Chat to library staff**
www.latrobe.edu.au/library - Access the text chat service from the Library website
- **Contact us**
www.latrobe.edu.au/library/about/contact - Call or email your Library

ASK LA TROBE

ASK La Trobe is a service open to all La Trobe students who need guidance with administrative matters during their time at La Trobe. ASK La Trobe can provide you with help relating to enrolments and timetabling, fees and financial support, exams, results and graduations & student life and support services.

Go to **ASK La Trobe – Current Students** at <http://latrobe-current.custhelp.com/app/home> and start by looking at our **FAQs** or visit us at your campus **ASK La Trobe Help Zone**.

If you can't find what you're looking for:

- **Call** 1300 La Trobe (1300 528 762)
- **Ask a question** online at <http://latrobe-current.custhelp.com/app/ask>
- **Visit the ASK La Trobe Help Zone**. Join the queue for the Bundoora, Bendigo or city campuses using the **QLess: Virtual Queuing** system.

LEARNING AND SUPPORT PROGRAMS

There are many resources, including services, to help you with your academic studies at La Trobe. Go to the **Learning and Support Programs** page at <http://www.latrobe.edu.au/students/study-resources/learning> for further information.

HEALTH AND WELLBEING

Your health and wellbeing are an important part of being a successful student. Studying can be stressful and many factors can affect your mental and physical health, in and outside of the University. Check out the services available to you at La Trobe to support your health and wellbeing at www.latrobe.edu.au/wellbeing.